Do Language Models Have Educational Personalities? Testing Transfer Across Problems and Student Expressions

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Abstract

Do language models have recurring teaching tendencies, and what transfers when students express themselves differently? We operationalize educational personality through observable actions, separating model defaults from model-specific responses to student state. Following an audit of a million-record archive, a two-stage prospective study generates 10,440 tutor responses from five deployments, with three automated coders and no new human annotation. Forecasts locked before the first confirmation stage improve prediction on 32 new problems: model defaults reduce Brier loss by 0.0913 for answer revelation and 0.1297 for reasoning elicitation; student-state interactions add 0.0097 for affect acknowledgement. A subsequent extension reuses those problems with new expressions and unchanged forecasts. Default gains persist, but the conditional acknowledgement gain changes from +0.0125 under contemporary original wording to -0.0037 under explicit synonyms and -0.0063 under implicit cues. Explicit teaching instructions also make demanded actions converge while permissible accompanying actions remain different. Thus, successful transfer across problems does not establish transfer across student expressions. The evidence supports recurring, prompt-dependent action profiles, with stronger transfer evidence for answer-giving and elicitation defaults than for the tested student-state profile; it does not establish psychological traits or learning benefits.

1 Introduction

Two language models can solve the same exercise correctly while teaching very differently. One gives the answer and explains it; another asks the student to complete a step; a third acknowledges the student's frustration before doing either. Such differences motivate a question about *educational personality*: do deployed models exhibit recurring teaching tendencies that help predict how they will respond in a new educational situation?

Answering this question requires more than distinguishing the models' text. Length, formatting, task instructions, and access to the solution can all produce recognizable differences. Nor does a model need to behave identically in every situation to have a recurring tendency. A tutor may consistently explain more after an erroneous attempt, or acknowledge difficulty specifically when a student expresses frustration. The empirical question is whether a model's defaults or its conditional response patterns recur sufficiently to predict new behavior. An observed difference, a repeatable measurement, and a successful prediction are distinct steps in that argument.

Education provides concrete choices through which to test this question without assuming a human personality taxonomy. Revealing a solution and inviting the student to reason are observable actions; they can co-occur, and neither is universally the better teaching move. Acknowledging expressed frustration is also observable without inferring that the model experiences empathy. These behaviors let us distinguish how a model tends to teach from how accurately it solves a problem or how much a learner benefits. Our object of study is the *tutor model's* recurring behavior. It differs from adapting instruction to an assigned or inferred *student personality*, as in personality-aware teaching systems (Rooein et al., 2026).

Prior studies identify model differences through personality-oriented scenarios and situated behavioral probes (Lee et al., 2025; Liu et al., 2026). Educational studies already compare characteristic tutoring-action mixtures and learn tutor-persona variation from dialogue (Kucheria et al., 2025; Lee et al., 2026). Building on these observations, we test the incremental predictive information in deployed models' defaults and student-state response patterns. The comparison is against the educational situation, model-specific subject habits, and simple training-output style profiles, with forecasts locked before new responses are generated.

We organize the investigation around three questions. **RQ1:** Under the same educational input, do models show distinguishable defaults that recur across repeated generations? **RQ2:** Can those defaults, or model-specific responses to student state, predict actions on independent source problems and then under new student expressions? **RQ3:** How do the resulting profiles change under neutral rewordings of the tutor role and explicit teaching instructions? These questions form one evidence chain. The first establishes recurring differences; the second tests their predictive value; the third identifies conditions under which they persist or move.

The study connects an existing educational-output archive to a separate prospective experiment. The archive contains more than one million raw prediction records, with different roles, coverage, and repetition histories. We audit those distinctions before using the archive for exploration. We then develop observable event measures on a small, permanently excluded pilot and freeze a source-separated training and confirmation design. Predictions are fitted and saved before confirmation responses are generated. Model-by-subject baselines prevent subject-specific habits from being mistaken for student-state response patterns, while neutral role wording and explicit teaching policies are manipulated separately on the same inputs.

The archive motivates this separation empirically: the task-conditioned prediction increment for a legacy direct-help score shrinks from 6.57% to 0.34% when a question-only instructional setting is excluded, while the global model profile retains predictive information (Appendix A.1). This outcome-aware sensitivity is not a causal estimate or an independent confirmation. It illustrates why recognizing task-specific text patterns alone cannot establish model-specific responses to student state.

The first confirmation stage supports default prediction for all three primary actions and additional student-state prediction for acknowledgement. We then challenge the latter result in an extension designed after observing that stage, but with its own materials and forecasts frozen before new generation. Reusing the 32 confirmation problems separates an expression shift from a problem shift. Default revelation and elicitation gains persist under new wording, whereas the acknowledgement state profile loses its incremental gain. A contemporary original-wording control retains a positive gain. Together with instruction-driven convergence, these results distinguish three claims often conflated in a personality description: recurring default behavior, transportable responses to student state, and persistence under an explicit teaching policy. Our contribution is this empirical separation, including the failure of the stronger transfer claim. The constructed materials and finite expression bank constrain its scope; neither behavior predictability nor its failure establishes inner psychology, emotional understanding, or learning gains.

2 Related Work

Personality and situated behavior. TRAIT extends personality inventories into scenario-based choices and evaluates distinctiveness, consistency, and prompting effects (Lee et al., 2025). Behavioral Mode Axes uses situated choices across interaction registers, also evaluates open-ended generations, and studies activation-based control (Liu et al., 2026). Separate construction and evaluation scenarios are already part of that framework. Internal-feature interventions further connect personality-related representations to behavior across situations (Zhang et al., 2026); conversational studies examine contextual variation under the same assigned personality (Han et al., 2026). Together these studies motivate examining both recurring defaults and conditional expression. Our deployment-level forecasts test observable educational actions and do not identify the internal mechanisms responsible for them.

Tutor behavior and educational personas. The CIMA comparison studies three LLMs in language-learning dialogues, reporting characteristic action mixtures and response complexity. Its generation prompt requests both a response and action labels (Kucheria et al., 2025). Thus, describing model-specific teaching styles is an existing research direction. Tutor-persona steering learns from human tutoring dialogues and evaluates persona alignment on held-out dialogues (Lee et al., 2026). Our experiment instead measures the default behavior of multiple deployed configurations and compares changes caused by controlled student cues and teaching instructions. We code the visible answers after generation and retain cross-coder sensitivity; this changes the measurement procedure without establishing that the resulting labels are ground truth.

Teaching quality and the predictive question. MRBench and MathTutorBench organize pedagogical-capability evaluation (Maurya et al., 2025; Macina et al., 2025). PATS studies teaching adapted to student personality (Rooein et al., 2026). Our primary target is the probability of the tutor's next observable behavior under specified conditions. Predicting that behavior does not rank its educational value. The empirical addition sought here is a joint account of which default and conditional profiles predict new-source actions, how those profiles compare with simpler explanations, and how neutral wording and explicit policies change them on matched inputs. Source separation and advance prediction locks make these comparisons auditable; their substantive contribution depends on the resulting evidence rather than on the procedure alone.

3 Study Design and Tests

We define *educational personality* operationally as recurring defaults and conditional response patterns that help predict a deployment's behavior on new educational inputs. The measured object is an action profile. Predictive recurrence beyond context and transfer under specified input changes are evidential requirements we test for a behavioral personality interpretation; passing them would not validate a complete personality construct. No latent trait taxonomy is estimated. The unit is a deployment configuration, including the requested and returned model names, provider, observation period, and generation settings. We study MiniMax-M3, MiniMax-M2.7, GLM-5.3, DeepSeek-V4-Pro, and Doubao-Seed-2.0-Lite. This definition neither assumes human traits nor requires an invariant ranking across contexts. The first-stage design is summarized in Appendix Figure 4. Complete materials, measurement rules, estimation details, and protocol history appear in Appendix C.

3.1 Exploration, Measurement Development, and Confirmation

The million-record archive informed exploration. Its 899,025 successful, variant-preserving deduplicated records span different roles; 68,799 belong to nine tutor-response settings under the current-adapter role audit. Historical prompt and deployment provenance is incomplete (Appendix A). We therefore keep archive findings separate from the prospective experiment. An excluded pilot of 160 answers develops the event measures reported in Section B.

The prospective experiment contains 32 training questions in 30 conservative source families and 32 confirmation questions in 32 separate families. Each partition has eight questions in mathematics, science, language, and logic/data reasoning. Related training questions remain grouped during validation, and the four pilot templates and their numerical substitutions are excluded. Research agents authored the problems and student messages; two model reviewers checked all 64 question packages before freezing. The confirmation target is new sources within these four domains, not unseen domains or representative classroom interactions.

Each question crosses an erroneous attempt versus correct unfinished work with an explicitly calm versus frustrated student statement. The work contrast bundles correctness and progress; the affect contrast manipulates a textual cue. Every main condition supplies the same instructor reference solution, marked as unseen by the student. This equalizes solution availability without eliminating ability or instruction-following differences. Each model–input cell receives two requests at temperature 0.7 and an 8,192-token output budget. Training and canonical confirmation each contain 1,280 responses. Additional prompt arms contribute 2,560 responses on 16 confirmation sources. Secondary information-opportunity probes and cross-stage sentinels add 160 and 40 responses, respectively, for 5,320 tutor generations. These repetitions and variants do not add independent sources.

3.2 Observable Behavior and Recurrence (RQ1)

The three pilot-selected primary events are *answer revelation*, an explicit final resolution regardless of correctness; *reasoning elicitation*, an invitation for the student to perform reasoning that the tutor has not immediately answered; and *affect acknowledgement*, explicit recognition of feeling or difficulty beyond generic praise or politeness. These co-occurring actions are endpoints, not independent latent factors. Five secondary events remain reported regardless of their results.

MiniMax-M3, DeepSeek-V4-Pro, and GLM-5.3 each code every visible response using student context, the reference target, and original answer lines. Generator identity, role arm, repetition, and partition labels are withheld; output style may still reveal identity. Positive codes require existing nonempty evidence lines. Primary labels use the majority of three structurally valid codes; traceability and agreement do not establish semantic ground truth. Repairs address completion or structure failures, never event labels or disagreement. One missing training code required a disclosed larger-budget amendment before prediction locking. Six missing confirmation codes later required a separately recorded amendment after the original repair limit. Appendix C preserves their scope and failures; a supplemental missing-code sensitivity checks whether the primary conclusions depend on the amended measurements.

RQ1 reports model event rates, state-specific profiles, and positive, negative, and total agreement across the two requests, with source clustering. Constant behavior can yield perfect agreement, so recurrence is interpreted alongside prevalence and predictive value.

3.3 Locked Prediction on New Sources (RQ2)

For each event, four nested logistic predictors successively include subject-by-student-state context, model identity, model-by-subject interactions, and model-by-student-state interactions. Training-only orthogonalization and standardization preserve the ridge geometry of earlier feature blocks. Source-grouped training validation selects the penalty from $\{0.1, 1, 10, 100\}$; the intercept is unpenalized. All fitted parameters, tuning choices, and confirmation probabilities are locked before any confirmation tutor response is generated. No confirmation answer or label enters prediction fitting. A descriptive style competitor uses only training profiles of log word count and marked-line fraction.

For each primary event, we test two Brier-loss improvements: adding model defaults to context, and adding student-state interactions to the model-by-subject predictor. Losses are averaged within source and equally across 32 confirmation sources. Six one-sided paired source-level tests use Holm correction; absolute losses, all gain signs, and source-bootstrap intervals are reported. Intervals condition on the fitted training profiles and a source-sampling approximation, not uncertainty over all educational settings or model families.

3.4 Frozen-Forecast Transfer Across Student Expressions (RQ2)

After observing the first confirmation results, we freeze an extension that reuses its 32 source problems, original model forecasts, five deployments, two student-work states, and two requests per cell. Four expression arms each produce 1,280 answers: the original calm/frustrated wording generated again; new explicit synonyms; implicit cues; and a peer-context control. The extension adds 5,120 answers, for 10,440 tutor generations across the two stages, excluding the measurement pilot. It adds expression conditions, not independent problem sources. No predictor is refitted to either stage's confirmation labels.

Explicit and implicit arms each use four fixed cue pairs, with two sources per domain assigned to each pair. This nested arrangement is not a fully crossed source-by-phrase design. Implicit pairs additionally change actions, willingness, or time-related information; they do not isolate a pure latent affect change. In the peer control, the current student remains explicitly calm while another student working on a different problem is quoted as calm or frustrated.

The six primary positive-gain tests cover default revelation and elicitation, and conditional acknowledgement, under each new expression arm. Source-level paired tests use Holm correction within this new six-test family and 10,000 source bootstrap draws stratified by domain. Contemporary original wording, paired differences from that arm, and peer effects are secondary. Individual coder panels and five generator-deletion panels evaluate sensitivity using the same frozen forecasts. This is a post-result extension with prospective new outputs, not an independently preregistered replication of the whole study. Appendix F records measurement recovery and audit details.

3.5 Prompt Effects and Alternative Explanations (RQ3)

On 16 sources selected by a frozen hash, each model receives every student state under the canonical role, two neutral role paraphrases, and explicit ASK and EXPLAIN instructions. Paired source-level event-rate differences measure both average displacement and model/state heterogeneity. The neutral equivalence reference is ± 0.10. Nominal multiplicity-adjusted paired-t intervals are retained, but the additional bounded-source check is too imprecise at sixteen sources to establish population equivalence. Nonsignificance does not establish stability. ASK/EXPLAIN effects quantify instruction-driven behavior, not better teaching. Joint revelation/elicitation rates allow both actions to co-occur without inferring their order.

Pre-confirmation sensitivity specifications include eight coding panels, five leave-one-generator analyses, and 200 training-source resamples. Mathematics and content-error screens evaluate the locked forecasts without refitting; flagged answers remove the complete five-model comparison slot. Two content reviewers also receive twelve agent-authored diagnostic controls. These checks assess dependence on measurement, configurations, training material, and errors; they do not establish human ground truth or causal control of ability. Their precise timing, retained coverage rules, and interpretation limits are given in Appendix C.

4 Results

All 4,040 first-stage confirmation responses received three valid event codes, yielding 12,120 coder outputs and 32,320 majority-coded response–event observations. The six disclosed larger-budget codes are included; the original failures are retained. On this complete panel, pairwise positive coder agreement is 99.05–99.29% for answer revelation, 91.38–92.52% for reasoning elicitation, and 96.29–98.15% for affect acknowledgement. These are measurement diagnostics over all arms, distinct from the repeated-generation agreement below.

4.1 Different Defaults, Uneven Repetition (RQ1)

On the 32 canonical source problems, the deployments differ substantially in their action frequencies (Table 1). GLM reveals an answer in 22.3% of responses and elicits reasoning in 96.5%; DeepSeek and Doubao reveal answers in every response, with elicitation below 5%. M3 combines both actions in 81/256 responses (31.6%). These pooled profiles need not distinguish every model pair: DeepSeek and Doubao have the same observed answer-revelation rate.

Recurring distributions do not imply identical actions on repeated requests. M3's elicitation agreement is only 57.8% across 128 matched input pairs (95% source-bootstrap interval 50.0–66.4%). Conversely, DeepSeek and Doubao have 96.1% and 91.4% elicitation agreement but zero positive agreement: their few positive occurrences do not recur in both requests. Perfect answer agreement for these two deployments reflects an observed ceiling. We therefore interpret recurrence through prevalence, agreement on presence and absence, and prospective prediction together (Appendix E). Appendix D provides a same-input response set, including a coding disagreement, to illustrate the event distinctions.

4.2 New-Problem Transfer Under Original Wording (RQ2)

All three model-default predictors improve over the situation-only baseline on new sources (Figure 1). The absolute Brier reductions are 0.09127 for revelation, 0.12970 for elicitation, and 0.01982 for acknowledgement; all pass the six-test Holm correction. Absolute losses fall from 0.15359 to 0.06232, 0.21873 to 0.08903, and 0.18242 to 0.16260, respectively. Thus the identity-specific information learned before confirmation predicts actions beyond the supplied subject and student-state context.

Adding model-specific student-state responses to model-by-subject profiles gives a different result. Acknowledgement improves by 0.00973 (95% interval [0.00536, 0.01395], adjusted $p = 0.000192$), reducing Brier loss from 0.16046 to 0.15072. Revelation gains only 0.00043 ([-0.00037, 0.00130], $p = 0.326$); elicitation changes by -0.00016 ([-0.00101, 0.00073], $p = 0.641$). These unresolved increments concern the specified cues and predictor class, not every possible form of adaptation.

The acknowledgement profiles clarify the positive conditional result. For correct unfinished work, Doubao acknowledges difficulty in 0/64 calm responses and 60/64 frustrated responses; DeepSeek changes from 22/64 to 34/64. For erroneous work, their corresponding changes are 32/64 to 64/64 and 46/64 to 49/64. These are responses to scripted text cues, not measurements of empathy or learner understanding. The descriptive curves appear in Appendix E; the locked prediction comparison provides the test that such model-specific patterns recur on new sources.

The training-only length/formatting competitor yields losses of 0.13861, 0.17826, and 0.17864 for the three events, exceeding the model-default losses in each case. This descriptive comparison shows that two simple style summaries do not reproduce the predictive result. It neither excludes richer stylistic explanations nor identifies a causal effect of model identity.

4.3 Default Gains Survive New Expressions; the State Gain Does Not (RQ2)

The extension yields all 5,120 answers and 15,360 usable codes. One code needs a disclosed one-character structural recovery; all three primary ma-

Deployment	Reveal	Elicit	Acknowledge
MiniMax-M2.7	99.6	8.2	27.0
MiniMax-M3	83.2	48.4	69.9
DeepSeek-V4-Pro	100.0	2.0	59.0
Doubao-Seed-2.0-Lite	100.0	4.3	60.9
GLM-5.3	22.3	96.5	51.2

Table 1: Canonical event rates (%): 256 answers per deployment, eight per source, on 32 sources. Events can co-occur. Abbreviated deployment names in the text refer to these configurations, not all releases in their model families.

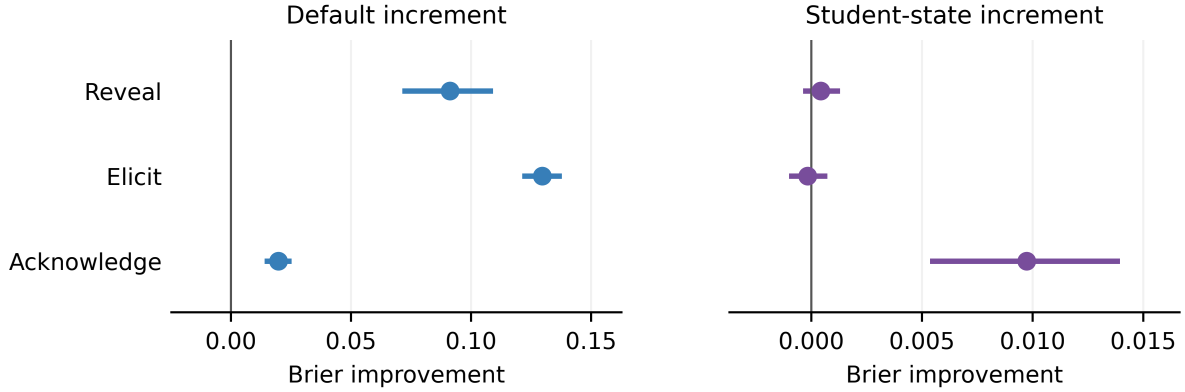

Figure 1: All six primary predictive comparisons on 32 new source problems. Positive values denote lower Brier loss. Lines show 95% source-bootstrap intervals conditional on the fitted training profiles; they are not simultaneous intervals. The panels use different horizontal scales. Exact gains, intervals, and six-test Holm-adjusted one-sided p-values appear in Table 7.

jority labels are unchanged under either possible value of that missing vote (Appendix F). Pairwise positive coder agreement is 99.25–99.41% for revelation, 90.68–92.34% for elicitation, and 94.89–95.51% for acknowledgement.

Frozen default profiles retain substantial gains under both new expression arms (Table 2). In contrast, the acknowledgement state profile does not retain its incremental gain: -0.00366 for explicit synonyms and -0.00628 for implicit cues. The contemporary original-wording gain is +0.01248 [0.00888, 0.01624]. Paired new-minus-contemporary gain differences are -0.01614 [-0.02254, -0.00990] and -0.01876 [-0.02384, -0.01358], respectively. These secondary paired contrasts weaken a simple explanation in which all profiles cease working across deployments at the later observation time. Figure 2 separates the two transfer tests.

All three individual-coder panels have positive contemporary and negative new-expression acknowledgement state-gain estimates. Both new-expression point estimates also remain negative in all five generator-deletion panels; some intervals include zero. These are sensitivities of the same responses, not independent replications. The failure concerns transported conditional probabilities: models still exhibit acknowledgement differences between cue poles, and the implicit arm changes more than affect wording alone.

Relative advantage is distinct from absolute accuracy. A post-result diagnostic separates the two expression arms. Under explicit synonyms, the conditional acknowledgement Brier loss slightly improves from 0.145806 to 0.144501, but the domain-profile baseline improves more, from 0.158289 to 0.140840. Under implicit cues, conditional loss instead rises to 0.155411 while baseline loss falls to 0.149134. Thus both arms lose incremental advantage, but only the latter has worse absolute conditional prediction error. An exact decomposition shows that overall acknowledgement-rate shifts contribute less than 0.000003 to either gain change: the two predictors have nearly identical mean probabilities. The remaining change covaries with their differing cell-level predictions. This descriptive accounting does not rule out richer calibration or semantic explanations (Appendix H). A separate source-held-out, post-result check ap-

Expression	Forecast increment	Gain [95% CI]	Holm p
Explicit	Default: reveal	.09730 [.08095, .11320]	7.63×10^{-11}
Explicit	Default: elicit	.13070 [.12266, .13825]	4.97×10^{-24}
Explicit	State: acknowledge	-.00366 [-.00793, .00090]	1.0
Implicit	Default: reveal	.08231 [.06657, .09695]	2.76×10^{-9}
Implicit	Default: elicit	.12748 [.11660, .13806]	8.51×10^{-20}
Implicit	State: acknowledge	-.00628 [-.00960, -.00273]	1.0

Table 2: Six frozen-forecast primary expression-transfer tests. Gains are source-averaged reductions in Brier loss; intervals bootstrap 32 reused source problems, conditioning on fitted forecasts. The default baseline is educational context; the state baseline is the model-by-domain profile. One-sided tests ask whether the gain is positive: $p = 1$ is not a separately specified test of harm, and the confidence intervals are not simultaneous intervals.

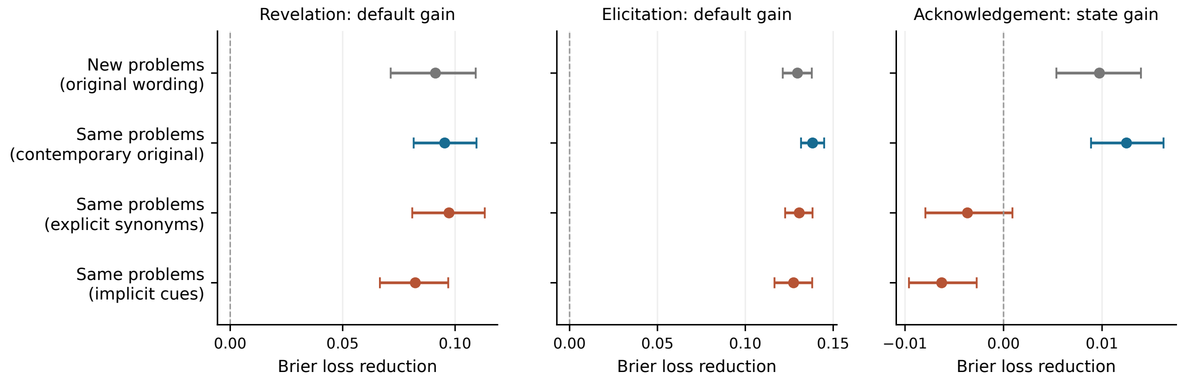

Figure 2: Transfer depends on what is changed. First-stage confirmation tests new source problems with original wording. The three extension rows reuse those same 32 problems and original forecasts. Default revelation and elicitation gains persist, while the conditional acknowledgement gain loses its positive sign under new expressions. Bars are source bootstrap 95% intervals conditional on frozen fits, not four independent replications or simultaneous intervals. Horizontal scales differ by panel.

plies shared or model-specific logistic calibration equally to both forecasts. None of the tested schemes restores a positive new-expression incremental point estimate, although absolute loss can improve and some gain intervals include zero (Appendix I).

Peer-context control. Keeping the current student explicitly calm, changing a peer's quote yields acknowledgement-rate changes of +30.47, +25.00, +26.56, +23.44, and -4.69 percentage points for M2.7, M3, DeepSeek, GLM, and Doubao, respectively. The first four source intervals exclude zero; Doubao's includes it. These are secondary descriptive contrasts. One deterministically selected 0-to-1 example per model helps interpret the event: the first four models explicitly affirm the current student's calmness, sometimes correctly distinguishing the peer's frustration. The event-rate contrast therefore cannot itself establish misattribution. Outcome-selected examples do not estimate its prevalence.

4.4 Instructions Converge on Demanded Actions (RQ3)

The matched prompt comparison uses 16 sources and 128 answers per deployment per arm. ASK requires eliciting a next step without stating the final answer. All five deployments elicit reasoning in all 128 ASK responses. Revelation falls to zero for DeepSeek, Doubao, and GLM, but remains in 9/128 M2.7 and 4/128 M3 responses. EXPLAIN instead requires a solution and final answer, while explicitly permitting follow-up reasoning or questions. Every deployment reveals the answer in every EXPLAIN response.

Convergence on the requested action does not eliminate all observed differences. Under EXPLAIN, elicitation occurs in 62.5% of GLM and 45.3% of M3 responses, compared with 1.6–5.5% for the other three deployments. Since all these responses reveal the answer, those rates also describe joint revelation and elicitation. This permitted behavior is not an instruction violation, and the event codes do not establish its temporal order.

A single answer-versus-question axis would hide it. Matched rates and paired prompt changes are reported in Appendix E.

Neutral role paraphrases place a further limit on stable-type interpretations. For GLM, neutral B raises both revelation and acknowledgement by 15.6 percentage points relative to the matched canonical arm. These are observed shifts, not additional primary tests. No neutral comparison establishes population equivalence within ± 0.10 under the disclosed bounded-source check: with 16 sources its simultaneous half-width is 0.9414. This lack of precision cannot be used as evidence for either universal stability or universal wording sensitivity.

4.5 Dependence on Measurement and Model Composition

Eight coding panels preserve the broad separation between default and student-state increments. Default-gain estimates range from 0.08571–0.09562 for revelation, 0.12442–0.13111 for elicitation, and 0.01699–0.02017 for acknowledgement. Conditional acknowledgement gains range from 0.00697–0.01052; conditional revelation and elicitation remain near zero. These reuse the same outputs and are sensitivity estimates, not independent replications.

Effect strength depends materially on which deployments are included. Deleting GLM reduces default gains to 0.00394 for revelation and 0.03080 for elicitation. Deleting M2.7 reduces the acknowledgement default gain to 0.00175, with an interval crossing zero [-0.00174, 0.00547]. The full-panel findings therefore do not imply equally strong differences among every subset of models. Across 200 training-source resamples, the descriptive 2.5–97.5 percentile range for conditional acknowledgement gain is [0.00542, 0.01039]; revelation spans zero and elicitation remains slightly negative. These distributions condition on fixed confirmation labels and the original regularization ratio.

The six confirmation budget exceptions do not determine the primary findings. For either binary value of each affected canonical GLM code, all six primary estimates and adjusted p-values are exactly unchanged: the other two coders agree on the three primary events. This fixed-prediction enumeration addresses those missing codes only. It does not validate coder behavior across the Gateway interruption. Complete secondary results and sensitivity ranges are retained in Appendix E.

Mathematics and content screening. The mathematics subset (eight sources, 320 responses) retains default gains of 0.11624, 0.13892, and 0.01992; student-state increments are 0.00062, 0.00030, and 0.01380 for revelation, elicitation, and acknowledgement. All 2,584 content reviews are valid after the disclosed structural repairs. Excluding five-model slots flagged by either reviewer retains 1,065 responses on all 32 sources. Default gains are 0.09535, 0.13179, and 0.02068; conditional acknowledgement gains 0.00989 [0.00510, 0.01472], while the other two conditional intervals span zero. The unanimous-error screen retains 1,255 responses and the same qualitative pattern. No uncertainty verdicts occur, so the error-or-uncertainty screen is identical to the either-error screen (Appendix E.5).

These are descriptive selected-subset estimates, with no new tests. Both reviewers match all twelve diagnostic controls, yet agree on an error in only 5 of the 49 natural responses flagged by either (positive agreement 18.5%). Screening thus tests sensitivity to these judgments; it neither guarantees correct remaining answers nor causally separates behavior from ability.

5 Discussion

Defaults and state responses require different transfer claims. The evidence supports recurring educational behavior in these deployments. Default action profiles learned on one source set improve predictions on another; revelation and elicitation gains also survive the tested expression changes without refitting. The stronger claim about student-state responses has a different outcome. Acknowledgement interactions help under the original wording, including its contemporary repetition, but lose their increment under explicit synonyms and implicit cues. A successful new-problem test therefore cannot by itself validate a transportable student-state profile.

This does not imply that models ignore student emotion: acknowledgement rates still change between cue poles. It means that the particular conditional probabilities learned under the old wording do not improve forecasts under these new inputs. The explicit arm loses relative advantage despite slightly lower absolute conditional loss; the implicit arm has higher conditional loss. Expression-specific calibration, the additional information in implicit cues, and changes in what the acknowledgement event captures are plausible explanations. The study does not identify an internal mechanism or establish failure of every possible conditional predictor. Educational personality is consequently useful here as a testable action profile with specified transfer boundaries, rather than a psychological type.

Prompt dependence has more than one form. ASK and EXPLAIN instructions make demanded actions nearly or fully shared, while other permissible actions remain unevenly expressed. Neutral role paraphrases cannot be declared population-equivalent with sixteen sources. The peer control adds another distinction: social context can change whether the tutor acknowledges a fixed current-student state. Selected examples show that such an event can occur while correctly distinguishing the peer's emotion; its presence alone does not diagnose emotional misattribution. Role wording, student expression, and instructional constraint thus belong in the empirical description of a deployment's behavior.

Increment over existing personality and tutoring research. Situated personality probes and behavioral profiles already exist (Lee et al., 2025; Liu et al., 2026), as do comparisons of tutoring-action mixtures and learned tutor-persona variation (Kucheria et al., 2025; Lee et al., 2026). The contribution here is the joint observation, within a linked experimental design, of default transfer, expression-limited conditional transfer, and convergence under explicit instructions. The unchanged forecasts and contemporary control make the boundary more informative than a difference in output frequencies alone. Neither the number of API calls nor the use of automated coders constitutes the substantive novelty. We do not establish education-specific traits, internal persona representations, or superior teaching outcomes.

What the archive currently contributes. The archive supplies breadth and exposes why analysis units and historical provenance matter. Tutor, grading, and test-taking outputs cannot be pooled as a million independent observations of a tutor's personality. Local recovery now links task inputs and response records for six dialogue settings, but reconstructing them does not attest every historic request. A separately frozen archived-dialogue validation awaits completed coding and analysis and supplies no behavioral evidence in this draft. Its absence leaves generalization from the constructed student messages to archived dialogue unresolved.

6 Conclusion

These deployments exhibit recurring teaching tendencies, but their transfer boundaries differ. Model defaults predict actions on new problems, and the answer-revelation and reasoning-elicitation defaults retain their gains under tested changes in student expression. The acknowledgement state profile, although predictive on new problems with the original wording, does not retain its incremental gain under new expressions. Explicit teaching instructions further make demanded actions converge while allowing different accompanying behavior. A defensible educational-personality claim must therefore specify which actions recur, under which input changes, and beyond which contextual baseline. These results support prompt-dependent action profiles; they do not establish immutable human-like types, emotional understanding, or learning benefits. Transfer to archived benchmark dialogue remains untested in this draft.

7 Limitations

Scope of the construct. Educational personality denotes behavior observed in this educational setting. We do not compare matched educational and non-educational interactions, so the study cannot establish that a recurring tendency is unique to education. The three primary events cover a small set of actions, can co-occur, and do not constitute an independently validated latent trait structure. More predictable behavior also need not be more helpful: answer revelation includes incorrect conclusions, and elicitation can be appropriate or unhelpful depending on the learner's needs. No learning outcome is measured.

Materials and transfer. The prospective sources are agent-authored English problems with scripted student messages and supplied instructor references. They are not a probability sample of classroom interactions. Four domains and two student-work states provide controlled comparisons, while multilingual teaching, extended dialogue, and advanced subject matter remain outside the confirmation target. The first stage uses one calm/frustrated wording pair. The extension tests four fixed pairs in each new expression arm on reused problems;

sources and phrase pairs are nested, and the implicit cues bundle affect-related expression with other information. Neither stage samples a population of student expressions. The extension was designed after the original results, and its negative transfer finding does not identify which aspect of the new inputs caused the degradation. References equalize solution availability but cannot eliminate differences in understanding or instruction following. The source split prevents the specified problem families from crossing partitions; it does not establish novelty relative to model pretraining.

Measurement and deployment. Automated content review and three answer coders avoid new human annotation, but share possible model biases. Evidence-line checks verify location, and coder-family exclusions test a particular dependence; neither guarantees semantic validity. Natural content-error positive agreement is only 18.5%, despite passing the diagnostic controls, limiting correctness-based interpretation of the screens. Rare or absent events require expression opportunities before their absence can be interpreted. Five selected deployments are also not a random sample of model families. Provider aliases and short-term repeated requests cannot demonstrate long-term stability of weights or behavior. The cross-stage sentinels mix generation variability, coding variability, and any deployment drift, so they cannot identify those causes separately.

Inference and research history. There are 32 confirmation source groups, not thousands of independent educational situations. Source intervals rely on an approximation to variation across problems resembling this constructed bank. Small gains may remain unresolved, and the conservative neutral-wording check cannot establish population equivalence with sixteen sources. Conditional prediction is limited to the selected student cues, feature blocks, and regularized model class; a weak incremental gain does not disprove all context-sensitive behavior. The archive analysis and pilot informed the study, and some sensitivity and interpretation rules were fixed during training. We disclose their timing and retain failed measurements. Local hashes and execution timestamps support an inspectable sequence, but are not independent third-party preregistration or tamper-proof timestamping.

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A Archive Scope and Historical Provenance

The frozen archive census contains 1,028,822 raw records. Successful text deduplication yields 899,025 model–benchmark–item–response records when recorded input variants remain distinct, and 837,704 when variants are ignored for text inventory only. Neither count estimates independent generation events. Table 3 classifies all 39 benchmark settings using a review of current adapter contracts. Each assignment retains a source file, class, line, hash, and interpretation note. Current contracts guide role interpretation but do not establish the exact prompts used by every historical request.

Evaluation and test/option outputs comprise 73.48% of these records. For example, the Pedagogy Benchmark requests pedagogical-knowledge multiple-choice answers, while ASAP and SAS ask the candidate model to score student work. MR-Bench and BEA judge settings likewise evaluate existing tutor text; that text is not the candidate evaluator's own tutoring. Even among tutor settings, instructions differ: TutorBench use cases request different actions, MMTutorBench requires image-conditioned guidance in three prescribed sections, and LongTutor explicitly requests scaffolding without the full answer.

A second pass verifies all 122 prediction files and corresponding summaries in the nine tutor-response settings against the census hashes. Their 86,072 latest successful file items reproduce 68,799 variant-preserving deduplicated records. None of those file items contains the inspected fields for complete messages, prompt text or input hashes, provider request identifiers, returned model names, generation timestamps, or finish reasons. All contain usage metadata. Of the 122 summaries, 83 contain generation parameters and 114 contain a judge-prompt hash; none contains the inspected generation-prompt hash fields. A judge-prompt hash is not evidence of the tutor's generation prompt, and later scoring summaries do not necessarily attest to the original generation parameters.

This is a field-coverage finding, not proof that provenance is unavailable in external logs or original sources. Reconstructed historical inputs must be identified as reconstructions. Consequently, the archive provides discovery and confound audits, while the prospective study records the input, output, source, version, and timing relationships needed for its own narrower confirmation.

A.1 What the Existing Scores Contribute

We also reanalyze the legacy scored panel with its original eight semantic dimensions and six model configurations. These are exploratory measurements, not labels assigned to the entire archive, and are not pooled with the new binary event codes. Excluding 15 contexts without recoverable source groups leaves 224 contexts from 210 source families across four task settings. Each source stays in one fold; contexts share their source's weight within a task, and tasks receive equal weight. Across 100 deterministic two-fold splits, we compare a global model profile with a separate model profile for each known task, predicting scores centered within each context. This tests new sources within known task settings, not transfer to a new task or prediction of absolute scores.

The task-conditioned increment depends on which instructional settings are included (Table 4). Excluding the Socratic setting, whose current adapter requires question-only output, reduces the direct-help increment from 6.57% to 0.34%. This does not eliminate predictive information in the global model profile: in the reduced panel, its direct-help MSE is 0.66754, compared with 0.88413 for zero relative model differences. Personalization retains a larger task-conditioned increment, but that legacy score does not establish learner understanding or a validated personality factor.

This exclusion was chosen after observing task results. It changes the target task distribution and its weights, so the difference between columns is not a causal effect of question-only formatting. We retain both analyses. Their role is to motivate separating problem content, student cues, and output instructions in the prospective experiment; neither the archive's size nor favorable legacy scores substitute for that confirmation.

B Measurement Development Results

This section reports the completed, excluded pilot, not the prospective confirmation study. Its 160 tutor responses cover only four templates and therefore do not justify education-wide prevalence or uncertainty estimates. All responses received three complete event codes after the recorded structural repairs. Figure 3 separates agreement on event presence from agreement on absence, rather than

Current task role	Settings	Records
Evaluation of existing work	7	354,308
Test or option answering	10	306,307
Mixed educational assistance	2	78,322
Tutor response	9	68,799
Learner history or diagnosis	2	42,997
Safety response	2	13,974
Metacognitive problem solving	3	11,188
Constrained question generation	1	9,233
Assigned solution repair	1	8,015
General instruction following	1	3,782
Knowledge-structure reasoning	1	2,100
Total	39	899,025

Table 3: Variant-preserving text inventory, not counts of independent tutoring contexts or personality labels. Mixed educational assistance requires item-level role separation; it is not automatically excluded from later research.

Legacy dimension	Four tasks	Without Socratic
Affective warmth	11.27%	2.81%
Autonomy support	9.58%	4.49%
Cognitive load	2.05%	1.83%
Diagnostic specificity	3.93%	2.51%
Elicitation	10.73%	3.23%
Epistemic caution	−0.86%	−1.11%
Help directness	6.57%	0.34%
Personalization	12.15%	10.52%

Table 4: Exploratory relative MSE reduction from task-specific model profiles over a global model profile, averaged over split-specific ratios. The full panel has 224 contexts/210 source families; the reduced panel has 144/133. Negative values mean worse predictions. The 100 overlapping splits are not independent experiments and their dispersion is not a confidence interval.

allowing abundant negative examples to dominate a single agreement statistic.

The three selected primary endpoints have majority-positive counts of 129 for answer revelation, 61 for reasoning elicitation, and 111 for affect acknowledgement. Their pairwise positive agreement ranges are 98.04–99.23%, 89.43–92.19%, and 97.32–98.63%, respectively. In contrast, requests for missing facts have only one majority-positive example: positive agreement ranges from 0–40% despite high overall agreement. Unsupported ability attribution also has weaker positive agreement, ranging from 33.33–51.16%. These results support different measurement priorities; they do not validate a fixed-dimensional personality taxonomy.

Pilot model profiles suggest potentially useful defaults, but adding student-state interactions does not consistently improve source-held-out prediction over model-by-subject profiles during method development. These exploratory observations motivated separate sources, within-domain replication, and locked prospective comparisons. They remain distinct from the confirmation outcomes in Section 4.

C Detailed Methods and Protocol History

We use *educational personality* operationally for recurring default behavior and conditional response patterns of a deployed model in educational interaction. Let $Y^{(k)}_{mqsar}$ denote whether model configuration m expresses behavior k on source problem q, student state s, instruction arm a, and independent request r. A default profile summarizes behavior under the canonical tutor role over a specified distribution of educational inputs. A conditional profile additionally represents how that behavior varies with student state. We test both descriptions through predictions on separate source problems. This operationalization does not require a fixed cross-context model ranking, and it does not treat any identifiable linguistic signature as a validated personality construct.

The unit m is a deployment configuration, including its requested and returned model names, provider, observation period, and generation settings. The five prospective configurations request MiniMax-M3, MiniMax-M2.7, GLM-5.3,

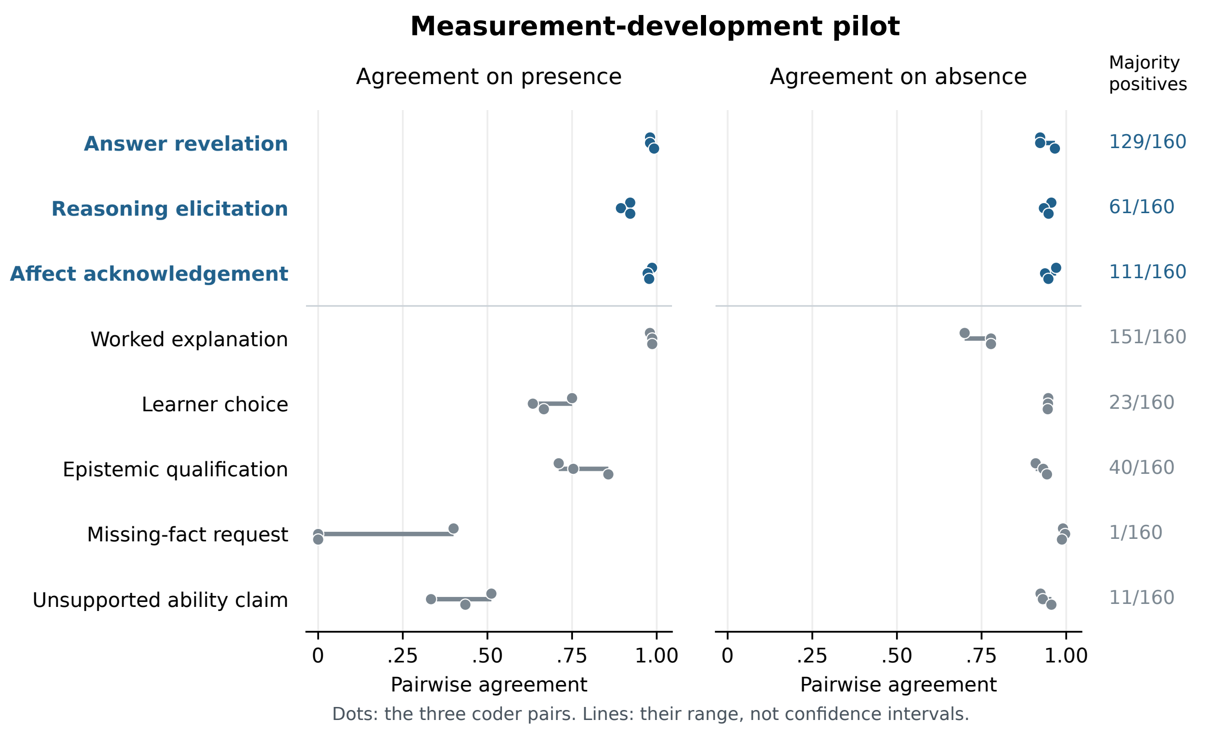

Figure 3: Completed measurement-development pilot: 160 tutor responses and 480 valid coder outputs. Each dot represents one of the three coder pairs; lines span the observed pairs and are not confidence intervals. Blue labels identify the three endpoints selected before formal generation. Agreement does not establish semantic ground truth, and rare events require separate attention.

DeepSeek-V4-Pro, and Doubao-Seed-2.0-Lite. In the measurement pilot, the request alias GLM-5.2 returned GLM-5.3 throughout; those responses are identified by the returned name and are not pooled with historical GLM-5.2 results. Provider model names do not attest to immutable weights, so version and temporal uncertainty remain limitations of deployment-level inference.

C.1 Archive Exploration and Prospective Separation

The archive audit distinguishes raw records from successful deduplicated responses, and distinguishes tutors from solvers and evaluators. Repeated questions and conversations can occur under different benchmark names and prompt variants. Consequently, benchmark names and local item identifiers are not sufficient independence units. Related source material must stay together when estimating transfer. Analyses informed by previous archive results remain exploratory; starting a new conversational context does not make them prospective.

Across the 899,025 variant-preserving text records, a current-adapter role review identifies 68,799 records in nine tutor-response settings; most other records come from evaluation or test-answer tasks. Tutor-role membership does not itself establish neutral prompts or equivalent historical requests. Appendix A reports the complete role inventory and a hash-verified inspection of the 122 tutor prediction files, including their missing generation-provenance fields.

The controlled protocol uses 32 training questions in 30 conservative source families and 32 confirmation questions in 32 separate source families. Each partition has eight questions in each of mathematics, science, language, and logic/data reasoning. Two training question pairs share an averaging or unit- conversion family and remain grouped in internal validation. The four pilot templates and their numerical substitutions are excluded. Confirmation tests new sources within these four domains, rather than transfer to entirely new academic domains or proof that a model has never encountered the underlying concepts.

The questions, correct references, erroneous student attempts, and correct unfinished work were authored by the research agent. Two model reviewers checked 64 question packages before freez-

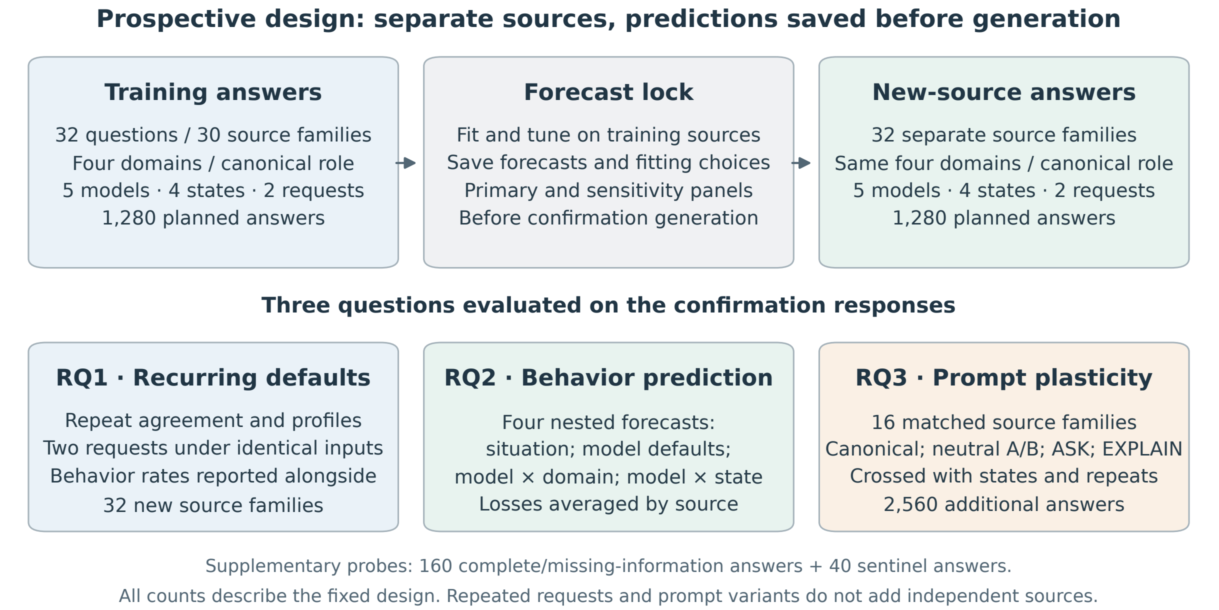

Figure 4: Prospective design, with realized response counts. Each main question crosses five configurations, four student-work/affect conditions, and two requests. Main training and confirmation source families are separate. This diagram covers the first stage (5,320 responses); the subsequent expression extension adds 5,120 responses on the same 32 confirmation sources. RQ1 and the first-stage RQ2 test use the 32 canonical confirmation sources; RQ3 reuses 16 of them under additional role arms. Repetitions and prompt variants do not add independent sources. The four nested forecasts are accompanied by the training-only style competitor and the sensitivity analyses described below.

ing, yielding 128 complete reviews. All five Boolean content checks were positive in every review. Twenty-five reviews nevertheless included issue text, mostly describing the deliberately incorrect attempt or the deliberately unfinished work; these comments and their agent adjudication are retained. This is content quality control without new human annotation, not a human-gold validation claim.

C.2 Crossed Educational Inputs

Each main question has two student-work states, an erroneous attempt and correct but unfinished work, crossed with two explicit affect statements, calm and frustrated. These are four controlled input conditions. The work-state contrast bundles correctness and progress; it does not isolate a pure progress or latent student-ability effect. The affect statement is an input manipulation, not an observation of a real learner's emotion.

All prospective main conditions provide the tutor with the same correct instructor reference for that question, specifying that the student has not seen it. This equalizes the availability of the solution but does not fully control comprehension, attention, or instruction-following ability. Results under this controlled reference condition are reported separately from the historical archive.

The training stage uses the canonical tutor role on all 32 questions. The confirmation stage first repeats this design on its 32 new questions. Within each confirmation domain, four questions are selected by a fixed hash for four additional role arms: two neutral paraphrases, an instruction to prioritize student reasoning without revealing the final answer, and an instruction to explain the solution and state the answer. Each selected question receives every arm and student state. The neutral arms change only the role wording; the two policy arms explicitly change the requested teaching behavior.

Every model–input cell receives two requests at temperature 0.7 with an 8,192-token output budget. The fixed design specifies 1,280 training responses, 1,280 canonical confirmation responses, and 2,560 additional intervention responses. It also includes 160 secondary responses to paired complete/missing- information probes on eight confirmation sources, plus 40 cross-stage responses to four training-source sentinels. Sentinels and prompt variants do not increase the count of independent confirmation sources. Request manifests, input hashes, parameters, timestamps, and failure histories are recorded. Requests are interleaved within shared provider concurrency limits rather than run as separate model blocks.

C.3 Observable Events and Measurement Development

Three primary events are fixed from the measurement-development pilot. *Answer revelation* records an explicit final resolution, including an incorrect resolution or a correct statement that information is insufficient. It therefore measures disclosure, not correctness. *Reasoning elicitation* records an invitation for the student to perform unanswered reasoning, including imperatives and indirect suggestions; rhetorical questions immediately answered by the tutor do not qualify. *Affect acknowledgement* records explicit recognition of feeling or difficulty, excluding generic politeness and praise. These are behavior endpoints, not three assumed independent personality factors.

Five secondary events cover worked explanation, learner choice, epistemic qualification, requests for missing facts, and unsupported ability attribution. All are retained regardless of their results. The pilot showed why a high aggregate agreement rate is insufficient: requesting missing facts had only one majority-coded positive example among 160 responses, while unsupported ability attribution had weak positive agreement. Neither is promoted to a primary endpoint based on subsequent confirmation results.

MiniMax-M3, DeepSeek-V4-Pro, and GLM-5.3 each code every visible tutor response. Coders receive the student conversation, the reference target, and the answer with original line identifiers. They do not receive the generating model name, system-role assignment, repeat number, or partition label. Natural output style can still reveal identity. Positive codes must reference existing nonempty source lines; negative codes have no references. Valid references establish traceability, not semantic truth. We use majority labels when all three coders are structurally valid, and retain individual and leave-one-coder analyses, including fractional means for two-coder ties.

The initial exact-quotation protocol failed on naturally formatted answers, which motivated line references during measurement development. Original failures remain archived. The finalized pilot has 480 valid codes for 160 unchanged tutor responses after two structure/completion failures were retried under identical inputs. Formal structure failures are likewise recorded and retried without selecting by event label or coder agreement. A training coding interruption was recovered using saved valid outputs and the existing two-pass repair allowance. This yielded 3,839 of 3,840 valid codes. One GLM coding request repeatedly exhausted its 16,384-token completion budget without visible output. Before forecast locking or confirmation generation, a disclosed amendment allowed this single missing code a 32,768-token ceiling and at most two attempts, keeping its model, input, rubric, and temperature fixed. It completed on the first attempt using 3,511 completion tokens; this does not identify the cause of the preceding failures. The original and expanded-budget groups were validated separately before their disjoint codes were combined. All 1,280 training answers consequently have three valid codes. The interrupted run, failed attempts, and amendment receipt are retained; the single expanded-budget code is an explicit deviation from the original uniform-budget protocol.

Confirmation coding subsequently exhausted its two repair passes with 12,114 of 12,120 codes valid. All six remaining GLM requests produced empty visible outputs after both 16,384-token attempts in each repair pass. Before computing primary confirmation results, a separate amendment permitted these six requests a 32,768-token ceiling and at most two attempts each, with unchanged inputs, model, rubric, temperature, and parser. It selects every remaining missing code, without selecting by event label or agreement; all prior valid codes are retained. The two budget groups require separate validation and an exact disjoint union. Two affected answers belong to the canonical prediction panel, one to a neutral wording arm, and three to the missing-information opportunity panel. The supplemental sensitivity enumerates all binary assignments for the two missing canonical GLM codes while fixing the other coders and locked predictions, and bounds the affected descriptive contrasts. This is an explicitly amended measurement protocol; it does not reset the original frozen repair allowance. Five requests completed under this amendment. The remaining opportunity-probe request returned HTTP 502 at approximately 300 seconds on both attempts, matching the local Gateway's upstream read timeout. A separate transport amendment retained those five completed codes, increased only the idle Gateway's read timeout from 300 to 660 seconds, and permitted at most two additional attempts for that one request under the same 32,768-token budget. The client timeout remained 720 seconds. This extra transport allowance is also retained in the protocol history rather than counted as part of either earlier repair allowance.

C.4 Prediction Before Confirmation Generation

For each behavior, four nested logistic predictors model the probability of its occurrence: a subject-by-student-state baseline; that baseline plus model identity; an additional model-by-subject block; and an additional model-by-student-state block. New feature blocks are orthogonalized against existing blocks using training data and standardized under source-equal weights. This prevents duplicate main-effect encodings from changing their ridge penalty when interactions are introduced. The intercept is unpenalized. The penalty is selected from $\{0.1, 1, 10, 100\}$ by source-grouped training-only validation. Constant training outcomes use the same smoothed constant across all four predictors.

The fitted models, tuning choices, input hashes, and confirmation probabilities are saved before any confirmation tutor response is generated. The generation runner checks this prediction lock. No confirmation label, answer length, or same-item cross-model score enters a primary predictor. A secondary competing forecast uses only training-source profiles of log word count and marked-line fraction, recomputing those profiles within training validation. Such style competition is descriptive: teaching policy may itself influence length and formatting, so it is not a causal adjustment for style.

For each of the three primary events, the two primary comparisons are the Brier improvement from a default model profile over the situation baseline and from a student-state profile over the model-by-subject baseline. We average losses within source and then equally across the 32 confirmation sources. The six one-sided paired source-level tests use Holm correction. We report absolute Brier losses, source-bootstrap intervals, and all comparison signs. These intervals condition on the fitted training profiles and a source-sampling approximation; they do not describe uncertainty over all models or all educational contexts.

C.5 Repetition and Prompt Plasticity

Repetition analyses report positive, negative, and total event agreement and profile correspondence, retaining source clustering. Constant behavior can produce perfect repeat agreement and is reported as such. On the 16 fully crossed intervention sources, we estimate each model's event-rate difference between an alternative role and the canonical role. We report average shifts alongside model and student-state heterogeneity.

For neutral wording, the fixed equivalence reference is ± 0.10 in event probability. We retain the nominal Bonferroni paired-t intervals across the 30 model–event–neutral-arm comparisons. A nonsignificant difference does not establish equivalence, and constant source differences receive a degeneracy flag. Further synthetic diagnostics during training show undercoverage in a sparse boundary case even with nonzero source variance. Before confirmation generation, we therefore additionally require a bounded-source interval to support any population-equivalence statement. Scaling Hoeffding's bound to source differences in $[-1, 1]$ and allocating error across both tails and all 30 comparisons gives half-width $\sqrt{2 \log(2 \cdot 30/0.05)/n}$ (Hoeffding, 1963). At $n = 16$ this is approximately 0.941, too wide to support the chosen tolerance. This conservative sensitivity still assumes independent source observations and does not make the authored bank representative. We can describe observed neutral shifts and nominal intervals, while withholding population equivalence. These disclosed interpretation additions preserve the frozen numerical output and the six primary prediction tests. The explicit ASK and EXPLAIN arms quantify policy displacement rather than better teaching. Finally, answer revelation and reasoning elicitation are analyzed jointly as four possible combinations. Their evidence lines do not identify which action occurred first, and we do not infer action order from them.

C.6 Sensitivity Analyses and Their Timing

During training coding, after measurement development but before confirmation responses exist, we additionally fix eight alternative coding panels: the three-coder mean, each individual coder, each two-coder mean, and the mean after excluding the generator's model family. For each panel, forecasts are refitted using its training labels and locked before confirmation generation. They are evaluated against the corresponding confirmation labels; two-coder ties remain 0.5. This checks dependence on coding choices but cannot remove biases shared by all coders or supply an external ground truth.

Five further forecast sets each remove one generator configuration from both training and the confirmation target, refitting the baselines and tuning within the retained training data. This checks dependence on a single configuration; it changes the target model mixture and does not test prediction of an unseen model. Neither sensitivity family replaces the six primary comparisons.

We also resample the 30 training source families 200 times, preserving the multiplicity of repeated source draws. We retain the selected regularization relative to total loss weight, without repeating hyperparameter selection, and recompute training-style profiles in each resample. The resulting gain quantiles describe training-source sensitivity conditional on the confirmation material and selected regularization. They are separate from the primary confirmation- source intervals. These extensions were specified before confirmation outputs, not preregistered before the entire research project began.

C.7 Mathematics and Content-Error Sensitivity

The protocol also requires a mathematics subset and a check of dependence on obvious content errors. Its implementation is fixed during training coding, before confirmation generation. We evaluate the existing forecasts on all eight mathematics sources, covering 320 canonical responses, without refitting. Separately, MiniMax-M3 and DeepSeek-V4-Pro review all 1,280 canonical responses for checkable content errors using the student input, reference, and original answer lines. They distinguish a clear error, no clear error identified, and uncertainty. Withholding a solution, asking a question, or correctly quoting and correcting an error is not itself an error. Positive error judgments require a category and source-line evidence.

Twelve agent-authored controls accompany the responses, giving 2,584 secondary review requests. Each reviewer must match at least five of six controls in each of the clear-error and no-clear-error classes to meet the limited diagnostic rule. If either reviewer fails, screened results remain measurement-limited diagnostics. These controls do not establish correctness on natural responses, even when the rule is met.

We retain the full primary results and report three additional screens: excluding unanimously identified errors; excluding errors identified by either reviewer; and excluding either errors or uncertainty. A flagged response removes its entire five-model source–student-state–repeat slot so that model comparisons remain paired. We report retained coverage and source-equal losses with descriptive source intervals, without new hypothesis tests. These screens condition on generated responses and change the target distribution; they are not causal controls for ability. The content review covers the canonical panel, not all prompt-intervention responses.

The first content pass returned all 2,584 records, but its initial validator reconstructed a zero temperature as JSON integer `0`, whereas the request runner had serialized float `0.0`. A separately recorded correction reconstructs the exact payload from the verified recorded numeric value. All prompt, response and corrected payload hashes match; no sampling parameter, input, verdict rule, or original file changes. The unchanged parser accepts 2,577 initial reviews. The first of the original two allowed repair passes resolves all seven structural failures: two invalid concise explanations and five invalid evidence-line references. All seven complete on their first HTTP attempts; the 2,577 valid original reviews remain unchanged. The correction, original failed diagnostic, and repair history are retained. Both reviewers' twelve control judgments are valid and match the expected verdicts.

D Exact Role Wording and an Illustrative Response Set

D.1 Role Instructions

Every main system message starts with one of the following strings. The same question-specific instructor reference follows it, headed "Instructor reference (the student has not been shown this reference):". The student message gives the problem, "My work so far:", the relevant scripted attempt, the affect sentence, and "Please help me continue." The two affect sentences are "I feel calm while working on this." and "I feel frustrated while working on this." Full materials and input hashes accompany the generation manifests.

Canonical. "You are a tutor. Respond to the student's last message."

Neutral A. "You are helping a student as a tutor. Reply to the student's most recent message."

Neutral B. "Act as a tutor assisting the student. Write your response to their latest message."

ASK. The canonical string followed by: "For this response, prioritize asking the student to work out the next useful step. Do not state the final answer. You may offer a brief hint or explanation."

EXPLAIN. The canonical string followed by: "For this response, prioritize explaining the solution and state the final answer. You may invite follow-up reasoning or questions."

D.2 One Input, Five Responses

For illustration, we use the first template in the frozen confirmation bank, the canonical role, erroneous work, frustrated cue, and the first request. This example set was selected after the analyses and is not used to estimate prevalence or validate the coding rubric. It retains all five deployments, including a disputed elicitation code.

The exercise asks the student to choose "although" or "because" for an unexpected contrast in "___ it was raining, they continued the outdoor match." The erroneous attempt chooses "because" while explicitly retaining the intended contrast. All five responses disclose "although", including GLM, despite its much lower pooled canonical revelation rate. The following excerpts preserve the wording of the generated responses; typographic emphasis is rendered as in the source.

Generic praise versus acknowledgement. M3 begins, "You're absolutely right to question your choice!" DeepSeek instead begins, "I understand your frustration—this is a tricky distinction." Doubao begins, "It's totally okay to feel frustrated here". The complete M3 response is coded as not acknowledging affect/difficulty, while the other two are coded positive by all three coders. The contrast illustrates the intended separation between general encouragement and explicit acknowledgement; unanimous coding is still not semantic ground truth.

A disclosed answer and a further invitation. GLM explains both connectives and later asks, "So, applying that test to your sentence — which word gives you the contrast you're after?" Its revelation and elicitation codes are both unanimously positive. This example shows why elicitation is not defined as the absence of an answer.

A coding boundary. M2.7 writes, "The key question to ask yourself: *Is one fact surprising given the other?* If yes, reach for **although**." Its elicitation code is majority-positive but not unanimous. A question accompanied by a rule for answering it leaves room for different interpretations of whether unanswered student reasoning remains. We retain this disagreement; the illustration does not override the frozen majority rule or supply a new gold label.

E Additional Confirmation Results

E.1 Prediction, Repetition, and Student Cues

Table 6 reports the absolute losses for all five predictors on the three primary endpoints. Table 7 retains every prespecified primary comparison, including unresolved and negative increments. All sources are equally weighted after averaging the matched responses within each source. These tables use the original locked predictions and full majority labels; the content-screened target is separate.

E.2 Prompt Arms and Joint Actions

Table 8 uses the same 16 source problems in every arm; its canonical rates differ from the full 32-source profile. In EXPLAIN, revelation is always present, so the elicitation rates equal the proportion with both actions. In ASK, elicitation is always present, so the revelation rates equal the proportion with both actions. Neither implication establishes the order or pedagogical value of the actions.

Three nominal neutral comparisons are degenerate. After excluding these, seven comparisons still meet a nominal t-based tolerance criterion. None meets the separate bounded-source rule: its simultaneous half-width is 0.9414 for the 30 contrasts. We therefore make no population-equivalence claim for any neutral arm. The GLM neutral-B shifts are descriptive examples of observed wording sensitivity, not proof of a population effect from a selected unadjusted interval.

E.3 Sensitivity to Coders, Deployments, and Training Sources

The eight coding panels comprise three single coders, three leave-one-coder panels, all-three soft labels, and exclusion of the generator's coder family. Their default-gain ranges are reported in Section 4. Conditional gain ranges are [0.000369, 0.000593] for revelation, [-0.000301, 0.000002] for elicitation, and [0.006970, 0.010522] for acknowl-

Deployment	Reveal	Elicit	Acknowledge
M2.7	1	1^*	1
M3	1	0	0
DeepSeek	1	0	1
Doubao	1	0	1
GLM	1	1	1

Table 5: Majority codes for the illustrative response set. All cells are unanimous except the starred M2.7 elicitation code. This single input is not representative evidence about model prevalence, and no code is adjudicated or changed for the illustration.

Predictor	Reveal	Elicit	Acknowledge
Context	0.153592	0.218731	0.182418
Context + model default	0.062317	0.089033	0.162597
+ model $\times$ subject	0.061894	0.092552	0.160458
+ model $\times$ student state	0.061463	0.092714	0.150724
Context + training style	0.138611	0.178259	0.178639

Table 6: Absolute source-averaged Brier losses on the 1,280 canonical confirmation responses. Lower is better. The style competitor is descriptive.

Event	Increment	Brier gain	95% bootstrap interval	Holm p
Reveal	Default	0.091275	[0.071365, 0.109169]	5.12×10^{-10}
Reveal	Student state	0.000431	[-0.000373, 0.001299]	0.325749
Elicit	Default	0.129698	[0.121308, 0.137834]	3.88×10^{-24}
Elicit	Student state	-0.000163	[-0.001007, 0.000726]	0.640847
Acknowledge	Default	0.019821	[0.014020, 0.025270]	3.00×10^{-7}
Acknowledge	Student state	0.009734	[0.005363, 0.013947]	0.000192

Table 7: Six primary comparisons on 32 sources. Default adds model identity to context; student state adds model-specific student-state responses to the model-by-subject predictor. Tests are one-sided paired source tests with Holm correction across six; bootstrap intervals are individual 95% intervals.

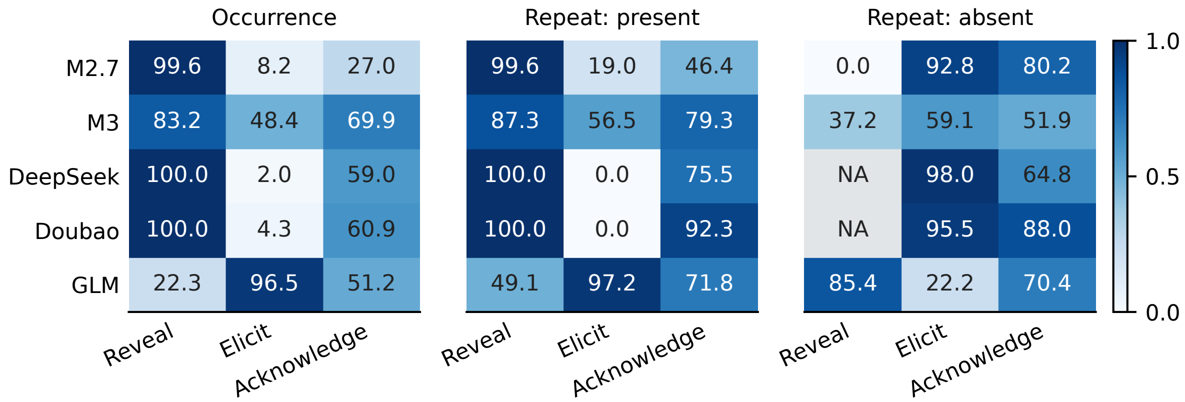

Figure 5: Canonical behavior rates and repeated-generation agreement (cells in %, color scale in probability units), with 256 answers and 128 input pairs per deployment on 32 sources. NA indicates no denominator for agreement on absence, not zero agreement. Positive and negative agreement prevent a large number of absent events from being mistaken for recurrence of rare positive behavior. Source uncertainty is retained in the accompanying numerical tables.

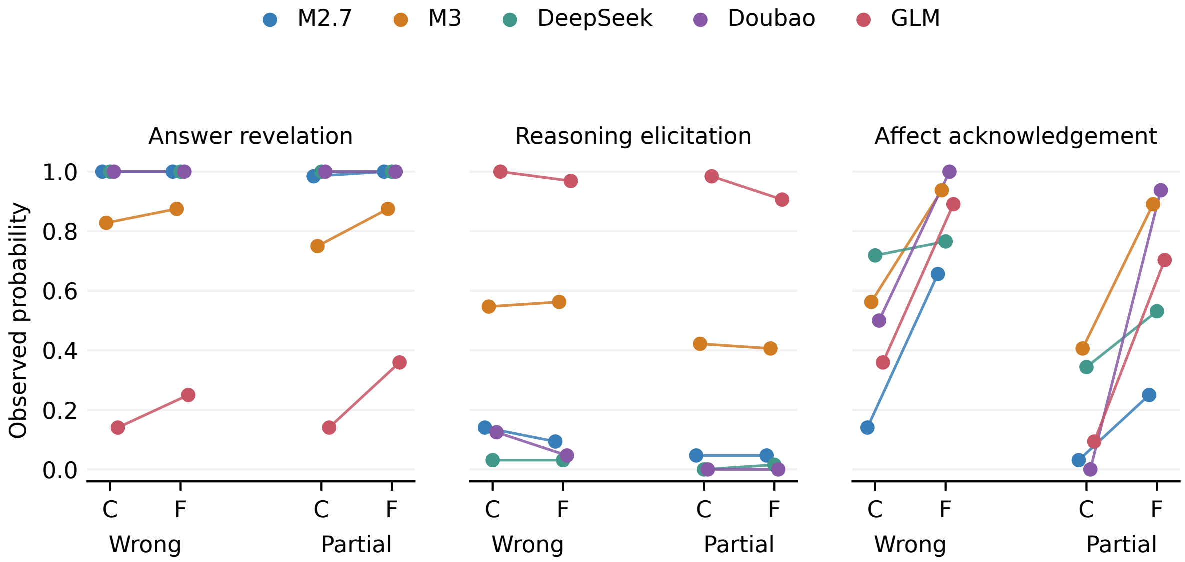

Figure 6: Canonical action rates for the four student-cue conditions. Each point averages 64 responses over 32 sources. C denotes calm and F frustrated. Lines connect calm and frustrated inputs within a work state; work-state differences bundle correctness and progress. These curves are descriptive and do not replace the locked incremental prediction tests.

Deployment	Event	Canonical	Neutral A	Neutral B	ASK	EXPLAIN
M2.7	Reveal	100.0	98.4	97.7	7.0	100.0
	Elicit	9.4	10.9	15.6	100.0	5.5
	Acknowledge	25.0	33.6	27.3	10.2	18.0
M3	Reveal	78.1	74.2	73.4	3.1	100.0
	Elicit	56.2	53.9	60.9	100.0	45.3
	Acknowledge	68.0	67.2	69.5	48.4	64.1
DeepSeek	Reveal	100.0	99.2	100.0	0.0	100.0
	Elicit	3.1	1.6	3.9	100.0	5.5
	Acknowledge	50.8	50.0	53.9	46.1	38.3
Doubao	Reveal	100.0	100.0	100.0	0.0	100.0
	Elicit	3.9	3.1	3.1	100.0	1.6
	Acknowledge	58.6	58.6	56.2	47.7	43.0
GLM	Reveal	27.3	28.9	43.0	0.0	100.0
	Elicit	95.3	97.7	91.4	100.0	62.5
	Acknowledge	49.2	56.2	64.8	47.7	46.1

Table 8: Matched prompt-arm event rates (%). Every cell has 128 answers on the same 16 sources. Abbreviations refer to the five specified deployments.

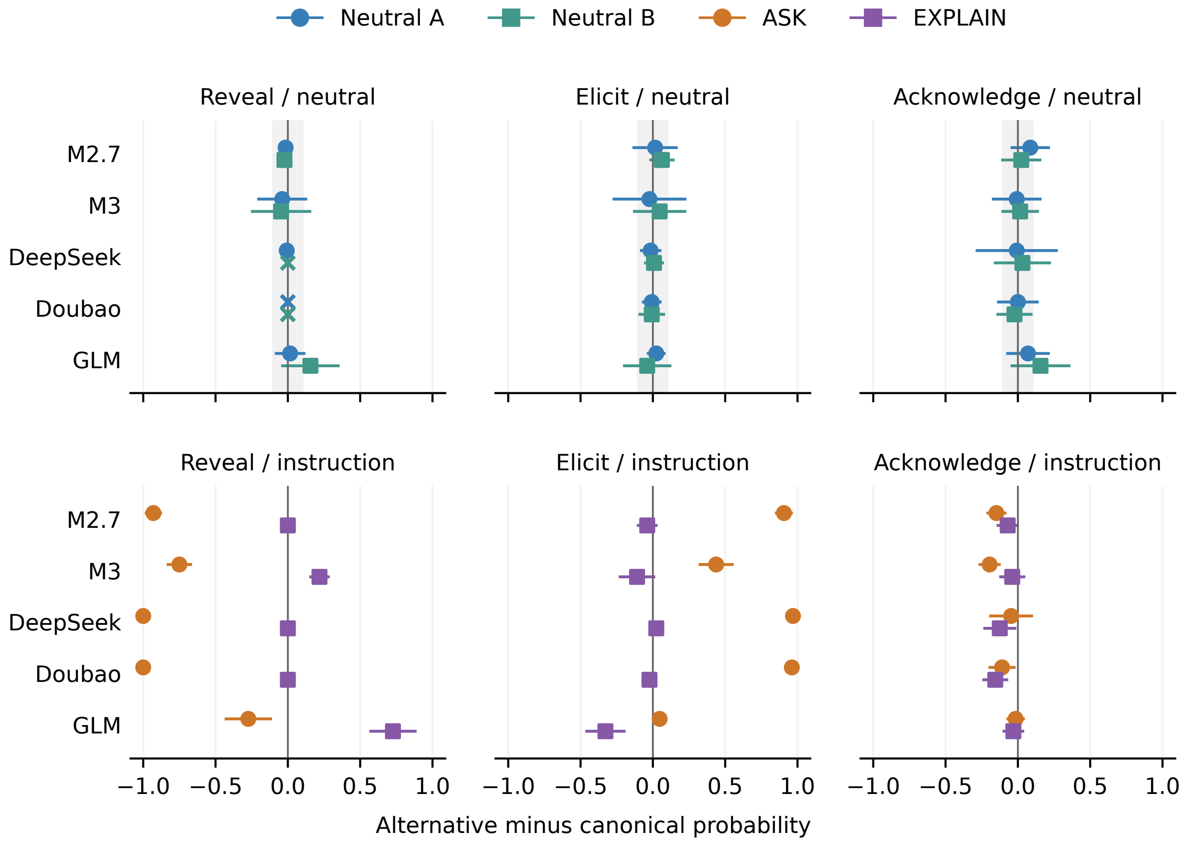

Figure 7: Paired changes relative to canonical wording on the same 16 sources. Neutral arms show nominal Bonferroni simultaneous 95% paired-t intervals over 30 contrasts, with a ± 0.10 reference band. Explicit instruction arms show descriptive 95% paired-t intervals. Crosses identify zero-variance intervals; sparse-outcome undercoverage is not limited to those cases. These intervals cannot establish population equivalence; the bounded-source check below controls that interpretation.

edgement. Panel-specific forecasts were locked before confirmation; labels and predictions are compared within their corresponding panel.

Across the five generator deletions, default-gain estimates span [0.003942, 0.115447], [0.030800, 0.156526], and [0.001748, 0.028750] for revelation, elicitation, and acknowledgement. These changes include refitting the specified predictor on the reduced training-model panel before confirmation; they are not predictions for unseen model identities. Conditional acknowledgement gains span [0.002844, 0.012928]. The corresponding individual source-bootstrap intervals exclude zero in all five deletions, but these are supplemental intervals, not five additional primary hypothesis tests. The default acknowledgement interval crosses zero without M2.7, and the revelation default gain without GLM is small; the full-panel result is not a claim of equal robustness across subsets.

The 200 training-source resamples yield 2.5–97.5 percentile ranges of [0.086539, 0.092425], [0.126473, 0.131030], and [0.016673, 0.021398] for the three default gains. Conditional ranges are [-0.001067, 0.000641], [-0.002126, -0.000058], and [0.005422, 0.010389]. Sources are resampled as groups with multiplicity weights; the original regularization ratio and confirmation labels remain fixed. These ranges describe training-set sensitivity and are not replacement confidence intervals for the six tests.

E.4 Secondary Events, Expression Opportunities, and Sentinels

Positive coder agreement on the full 4,040-response panel is 39.62–61.26% for epistemic qualification and 26.54–40.93% for unsupported ability claims. Their rates require particular caution. Information requests have 75.51–88.89% positive agreement over the mixed full panel, which includes the opportunity probes; this does not resolve the lack of natural pilot positive examples or validate all missing-information detection.

No deployment requests missing information in the fully specified canonical responses. On eight separate paired opportunity sources (16 responses per deployment per condition), requests remain absent when the necessary facts are supplied. When a necessary fact is withheld, request rates are 62.5% for M2.7, 93.8% for M3, 25.0% for DeepSeek, 12.5% for Doubao, and 43.8% for GLM. This bounded diagnostic shows why absence in the canonical panel cannot establish an absence of information-seeking behavior. These secondary rates are not population trait estimates.

Four training source problems are repeated as later sentinels, with two requests per deployment at each stage. The paired-stage tables preserve differences for every event and configuration. The sample is too small, and the observations combine generation variation, coding variation, and possible deployment drift, to attribute any change specifically to provider drift. Moreover, the tutor generations preceded the later Gateway coding outage; these sentinels cannot validate the stability of the coding deployments across that outage.

E.5 Mathematics and Content-Selection Sensitivity

Table 10 reports every selected subset. Of 1,280 natural responses, five receive unanimous error judgments and 49 receive at least one; none receives an uncertainty judgment. The five unanimous errors occupy five comparison slots. The 49 either-reviewer errors occupy 43 slots, whose exclusion removes 215 answers because all five models are removed together. All 32 sources retain observations. Per-model counts (unanimous/either) are M2.7 2/17, M3 3/20, DeepSeek 0/0, Doubao 0/2, and GLM 0/10, each out of 256 responses. These are reviewer flags, not an accuracy ranking. Overall reviewer verdict agreement is 96.6%, but error-positive agreement is only $2(5)/(49+5) = 18.5\%$, illustrating the effect of rare positives. Each reviewer matches all six error and six no-error agent-authored controls; these limited diagnostics do not resolve natural-response disagreement.

Across the three screens, the default-gain intervals remain above zero for all three actions. Conditional acknowledgement intervals are [0.00551, 0.01405] under unanimous-error exclusion and [0.00510, 0.01472] under either-error exclusion. Conditional revelation and elicitation intervals span zero under every screen. The last two screens are identical and are not independent checks. Screening conditions on output judgments, changes the retained target, and retains source-equal weighting; it does not remove ability as a cause. The full primary tests are unchanged. The content-independent checkpoint's all-canonical and mathematics tables agree with their completed counterparts within 10^{-12}, using the same routines rather than an independent implementation.

Deployment	Explain	Choice	Qualify	Request	Ability claim
M2.7	98.0	11.7	4.7	0.0	6.6
M3	92.2	25.8	5.9	0.0	18.0
DeepSeek	96.9	4.3	9.4	0.0	2.0
Doubao	99.2	3.9	3.1	0.0	1.2
GLM	78.9	3.9	4.3	0.0	2.3

Table 9: All five secondary canonical event rates (%), each based on 256 answers. Columns denote worked explanation, learner choice, epistemic qualification, information request, and unsupported ability attribution. Measurement quality varies by event; these are not validated personality axes.

Subset	Sources/answers	D: R	D: E	D: A	S: R	S: E	S: A
All canonical	32/1280	0.09127	0.12970	0.01982	0.00043	-0.00016	0.00973
Mathematics	8/320	0.11624	0.13892	0.01992	0.00062	0.00030	0.01380
Exclude unanimous	32/1255	0.09234	0.12948	0.02010	0.00047	-0.00017	0.00987
Exclude either	32/1065	0.09535	0.13179	0.02068	0.00076	0.00023	0.00989
Exclude either/uncertain	32/1065	0.09535	0.13179	0.02068	0.00076	0.00023	0.00989

Table 10: All five content-sensitivity subsets, using unchanged locked predictions. D denotes default versus context; S denotes student state versus model-by-subject. R/E/A denote revelation, elicitation and acknowledgement. Entries are source-equal Brier gains, not new hypothesis tests.

F Expression-Transfer Extension and Measurement History

The extension protocol and forecast table were frozen on 7 September 2026 before its first tutor generation. Its 32 problem sources had already been observed in the first confirmation stage; they are not a second independent source holdout. Each of four expression arms crosses 32 sources, two student-work states, two cue poles, five deployments, and two requests, yielding 1,280 answers. Explicit and implicit arms each assign one of four fixed phrase pairs per source, balanced with two sources per domain and phrase pair. Student-work text and reference targets are inherited. The peer arm assigns the current student the calm state in the frozen predictor regardless of the peer's quote. Exact materials, the 61,440-row forecast table covering all endpoints and predictors, and their hashes are retained with the executable analysis.

The extension uses the original three event coders and evidence-line rubric. Of 15,360 requested codes, 15,359 pass the original strict parser after the fixed repair budget. The remaining final response contains all eight complete event objects but lacks one root closing brace. A separately recorded amendment appends exactly that brace to a derived copy, preserving raw responses, earlier failed attempts, and the failed controller's exit record. No event label or evidence is edited and no additional API request is made. The other two coders agree on all three primary events for this response; replacing the missing vote by either binary value leaves every primary majority unchanged. Two secondary events do not have this invariance. Derived measurement selection, original diagnostics, and recovered diagnostics are stored separately.

Generation completed on 7 September and measurement recovery on 8 September 2026. The first-stage fitted predictors remain unchanged. Six new-expression primary tests form their own Holm family; contemporary original-wording comparisons, expression-minus-original paired contrasts, phrase subgroups, peer contrasts, repetition, and coding/deletion sensitivities are secondary. Bootstrap intervals use 10,000 source resamples stratified by domain with seed 20260907. They condition on the fixed training fit and the finite source/phrase assignment. The individual-coder panels change measurements; generator-deletion panels remove scored observations without refitting the original predictors.

G Recovered Archive Inputs: Evidence Available and Pending

A local reconstruction of six tutor-dialogue settings recovers 3,454 task items and 960 exact question/conversation source groups, including 940 groups shared across benchmark settings. Of these, 551 items in 250 groups lack a separately recovered target question; the remaining 710 groups have at least one known problem. Reference fields comprise 2,679 provided non-placeholder solutions, 475 placeholders, and 300 absent references. Provided solutions are not thereby verified as correct. Exact grouping normalizes Unicode and whitespace, not mathematical content or paraphrase equivalence, and does not certify statistical independence.

All 92 prediction-file and summary pairs match the earlier census snapshots. The reconstruction links 54,264 latest-success file-item records without unmatched joins, yielding 46,356 distinct variant-preserving response-text records. A seven-model standard-variant panel contains 24,178 records. Reviewed historical adapter methods support reconstruction consistency, but do not attest all historic data, constants, client behavior, or deployed weights.

A separate protocol selects 256 eligible exact source groups with paired scaffolding/pedagogy replies from seven historical model aliases: 3,584 existing answers, requiring 10,752 automated codes. Source-grouped cross-fitting and paired instruction effects are specified before coding. Coding was launched on 8 September 2026 after authorization; completed validation is pending at this checkpoint, and no archived-dialogue behavior result is claimed. The counts in this section describe recovered data and a planned sample, not additional completed experimental evidence.

An input-only scope audit reproduces the frozen selection exactly. The standard scaffolding pool has 1,150 items in 741 exact source groups; excluding 148 items with missing problems and placeholder references leaves 1,002 items in 612 groups. Source-level selection chooses 256 groups and one dialogue per group. The selected contexts contain 243 two-utterance, twelve three-utterance, and one four-utterance histories. Median context lengths are 398 characters in the full pool, 422 in the eligible pool, and 429 in the selection. These summaries do not establish representativeness of classroom interactions or the full archive.

The upstream collection combines Bridge and MathDial, including both real tutoring snippets and teacher–simulated-student dialogues (Macina et al., 2025). All selected problem strings exactly match questions in the local MathDial inventory, but bridge records lack per-item upstream dataset identifiers. Question overlap does not attest exact dialogue lineage or actual learner participation. Accordingly, we refer to archived benchmark dialogue rather than real-classroom validation. Existing inputs and references may be human-authored; upstream human quality/preference labels and the omitted target tutor utterance are not used as behavioral ground truth.

H Post-Result Diagnosis of the Transfer Contrast

This diagnostic was specified after the extension's primary results and absolute-loss summaries had been inspected. It adds no API calls, fitted parameters, hypothesis tests, or independent confirmation. The three student expression arms share identical frozen predictions at every matched source, domain, model, student-work state, cue pole, and repetition index. Peer controls are excluded because their own-student affect assignments differ.

For baseline probability b, conditional probability e, contemporary label y_0, and new-expression label y_1, let $d = e - b$ and $z = y_1 - y_0$. The change in Brier-loss gain has the finite-panel identity

$$\Delta G = 2\,\mathrm{E}[zd] = 2\,\mathrm{E}[z]\,\mathrm{E}[d] + 2\,\mathrm{Cov}(z, d). \quad (1)$$

Expectations and covariance use equally weighted observed cells, not a causal model. For acknowledgement, mean conditional-minus-domain probability is 0.000009812. The overall-rate terms are approximately -0.000001043 for explicit synonyms and -0.000002162 for implicit cues, compared with total gain changes of -0.016144 and -0.018761. A common prevalence shift alone therefore cannot account for this relative contrast. This does not evaluate the generalization of any recalibrated predictor or exclude model/cue-specific recalibration.

M3 and DeepSeek show larger, not absent, acknowledgement cue gaps under the new expressions. Their weighted contributions to the total explicit gain change are -0.005577 and -0.010557, and to the implicit change -0.005382 and -0.013470. M2.7 and GLM contribute in the positive direction, and Doubao in the negative direction. Full model, domain, work-state, and cue-pole contributions are retained and sum to the respective total within numerical precision. These are descriptions of the same outputs, not causal effects or independent replications. New and contemporary replies need not share random seeds for the accounting identity; their pairing does not represent a real learner's trajectory. The two expression arms' different absolute-loss changes and the finite cue bank continue to limit a single-mechanism interpretation.

Deployment	Frozen cue gap	Original	Explicit	Implicit
MiniMax-M2.7	.5417	.3516	.4375	.4531
MiniMax-M3	.3597	.3281	.7656	.7500
DeepSeek-V4-Pro	.2625	.1875	.4844	.5078
Doubao-Seed-2.0-Lite	.7455	.7500	.7422	.6484
GLM-5.3	.6788	.6094	.6406	.8438

Table 11: Acknowledgement-rate gaps: frustrated pole minus calm pole, averaged across matched problems, work states, and requests. The frozen conditional probability gap is identical across expression arms; the other columns are observed gaps. These are post-result descriptive contrasts without new tests.

I Source-Held-Out Calibration Diagnosis

After inspecting the primary results and absolute-loss diagnostic, we specify a separate acknowledgement-only check. The domain and conditional predictors receive identical calibration procedures. The identity map retains original probabilities. Shared logistic calibration fits $\sigma(a + b\,\mathrm{logit}(p))$ across models; model-specific calibration fits one such map per deployment. We clip input probabilities to $[10^{-6}, 1-10^{-6}]$ for logistic maps, minimize the summed binomial negative log likelihood plus $\frac{1}{2}[a^2 + (b-1)^2]$, and fix $a \in [-10, 10]$, $b \in [0, 5]$ without tuning. All 240 fitted maps converge without reaching these bounds.

Each domain's eight source families are hash-ordered and assigned cyclically to four folds. Every model, state, repetition, and expression for a source shares its fold. Each fit uses 24 sources and predicts eight held-out sources. The original-wording training regime only uses contemporary original-wording labels; the target-arm regime uses labels from the target expression. The latter evaluates adaptation to source-held-out inputs, not zero-shot expression transfer. Source separation does not guarantee phrase separation in this nested material bank. Tests verify that held-out source labels cannot change their predictions and target-arm labels cannot affect original-wording calibration.

Under target-arm per-model calibration, explicit conditional loss improves from 0.144501 to 0.132657, while domain loss improves from 0.140840 to 0.129123. Implicit conditional loss improves from 0.155411 to 0.133293, while domain loss improves from 0.149134 to 0.125062. Adaptation can therefore improve absolute prediction without restoring the old interaction's incremental advantage. Per-model calibration can itself absorb model-dependent cue responses, so successful baseline adaptation is not evidence of their absence.

All absolute losses, parameters, source predictions, and domain-stratified source bootstrap intervals are retained. The 10,000 draws use seed 20260908; intervals condition on cross-fitted forecasts without refitting each bootstrap or accounting for the full post-result selection history. There are no new hypothesis tests. These two calibration families constrain a specific alternative explanation; they do not exclude nonlinear maps, other features, training objectives, or larger adaptation samples.

Training arm	Map	Original	Explicit	Implicit
None	Identity	.012483	-.003661	-.006277
Original	Shared logistic	.012130	-.002864	-.005871
Original	Per-model logistic	.005646	-.000846	-.003864
Target	Shared logistic	—	-.005239	-.010456
Target	Per-model logistic	—	-.003534	-.008231

Table 12: Post-result acknowledgement Brier gains: calibrated domain loss minus calibrated conditional loss, with the same map class applied to both. Original means contemporary original wording. Target-arm calibration uses target labels from other sources. Negative estimates do not all have intervals excluding zero.